

Roll No.:

Total No. of Questions : 4]

[Total No. of Printed Pages : 11

11thARF(S/Z)JD-2026

1524-C

CHEMISTRY

Time : 3 Hours]

[Maximum Marks : 70

Note:- “Attempt any 60 Marks out of 70 Marks”.

SECTION-A

1. Select the correct one: (1 mark each)

(i) Pure substances are :

(a) Homogeneous mixture

(b) Heterogeneous mixture

☒ (c) Both (a) and (b)

(d) None of these

- (ii) 4.6×10^{22} atoms of an element weigh 13.8 gram. The atomic mass of the element is :

(a) 290

☒ (b) 180

(c) 34.4

(d) 10.4

- (iii) The nucleus of an atom consists of

☒ (a) Protons and Neutrons

(b) Protons and electrons

(c) Neutrons and electrons

(d) Protons, Neutrons and electrons

- (iv) Which one of the following is an Amphoteric oxide?

(a) MgO

☒ (b) Al_2O_3

(c) SiO_2

(d) P_2O_5

(v) Which one of the following is a state property?

(a) Heat

(b) Work

☒ (c) Internal energy

(d) Potential energy

(vi) In the evaporation of water, the entropy

(a) Decreases

☒ (b) Increases

(c) Does not change

(d) Sometimes increases, Sometimes decreases

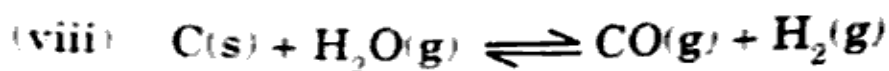
(vii) The state of equilibrium refers to

(a) State of rest

☒ (b) Dynamic state

(c) Stationary state

(d) State of inertness



the above equilibrium when subjected to pressure

(a) Remains unaffected

☒ (b) Proceeds in the backward direction

(c) Proceeds in the forward direction

(d) None of the above

(ix) The non-metal not undergoing disproportionation reaction is

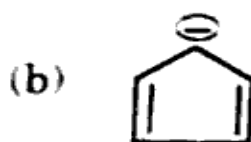
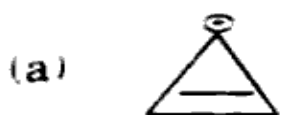
(a) P_4

(b) Cl_2

(c) I_2

☒ (d) F

(x) Which one of the following is Aromatic



☒ (d) All the above

SECTION-B

(Very Short Answer Type Questions)

(2 marks each)

Answer the following questions :

- (i) Define law of reciprocal proportions. Explain with a suitable example.
- (ii) Why do elements in the same group have similar physical and chemical properties?
- (iii) Draw the Lewis structures for the following molecules and ions BeF_2 and CO_3^{2-}
- (iv) Which out of NH_3 and NF_3 has higher Dipole moment and why?

- (v) Calculate the number of kJ of heat necessary to raise temperature of 60.0g of aluminium from 35°C to 55°C. Molar heat capacity of Al is $24 \text{ J mol}^{-1} \text{ K}^{-1}$.
- (vi) Define entropy change and write units of entropy change.
- (vii) Predict if the solutions of the following salts are neutral, acidic or basic: NaCl, NaCN, NH_4NO_3 and KF.
- (viii) Write IUPAC names of the products obtained by the ozonolysis of the following compounds.
- Pent-2-ene
 - 1-phenylbut-1-ene
- (ix) What is the difference between distillation and distillation under reduced pressure?

SECTION-C**(Short Answer Type Questions)**

3 marks each.

Answer the following questions

- (i) Define bond energy. How it is related to Bond dissociation energy.
- (ii) Calculate the mass percent of different elements present in sodium sulphate (Na_2SO_4)
- (iii) Explain, give reasons, which of the following sets of quantum numbers are not possible
- (a) $n = 1, l = 0, m = 0, s = +1$
- (b) $n = 2, l = 0, m = -1, s = +\frac{1}{2}$
- (c) $n = 2, l = 2, m = +1, s = -\frac{1}{2}$
- (d) $n = 2, l = 1, m = 0, s = -\frac{1}{2}$

(e) $n = 0, l = 1, m = 0, s = +\frac{1}{2}$

(f) $n = 3, l = 1, m = 0, s = -\frac{1}{2}$

(iv) Define electron affinity. What are its units. List three differences between electron affinity and Electro negativity <https://www.jkboseonline.com>

(v) Explain the concept of Resonance with suitable examples and define Resonance energy.

(vi) Define the following

(a) pH of a solution

(b) Solubility product

(c) Common ion effect

(vii) Discuss any three applications of redox reactions.

(viii) Discuss the chemistry of lassaigue's test.

- (ix) Why is Wurtz reaction not preferred for the preparation of alkanes containing odd number of carbon atoms. Illustrate your answer by taking one example.

SECTION-D

(Long Answer Type Questions)

(5 marks each)

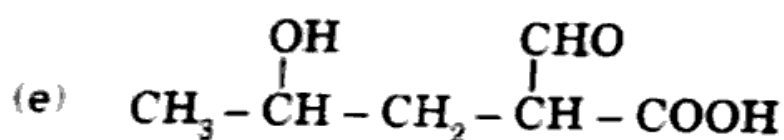
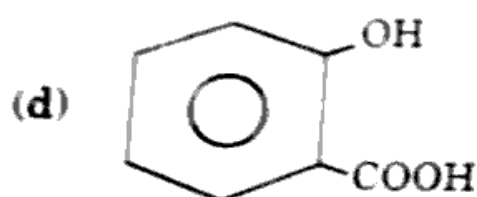
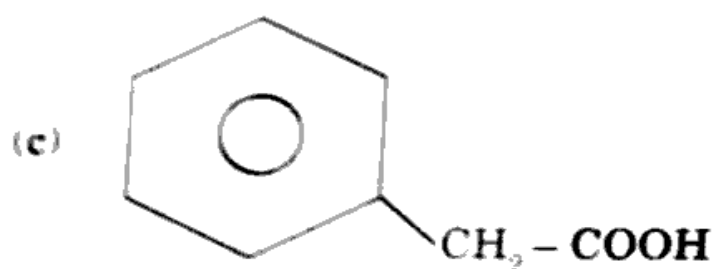
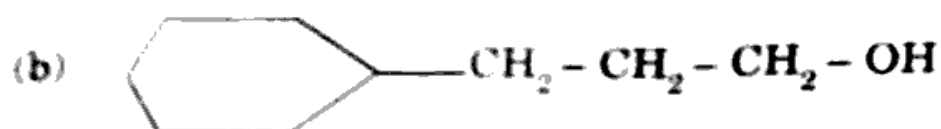
Answer the following questions

- (i) (a) What is electromagnetic spectrum?
- (b) Explain Absorption spectrum.
- (c) How will you show that the spectral lines are the finger prints of the elements

OR

Write the main postulates of Bohr's theory. Explain the quantisation of energy. What are the limitations of Bohr's theory?

(ii) Write the IUPAC names of the following compounds.



Or

What are free radicals and carbanions? How are they generated? Explain the relative stability of alkyl free radicals and alkyl carbanions.

(iii) Write notes on

- (a) Wurtz reaction
- (b) Corey-House alkane synthesis
- (c) Huckel's rule

Or

How will you prepare the following compounds from benzene?

- (a) Chlorobenzene
- (b) Nitrobenzene
- (c) Toluene
- (d) Benzene Sulphonic Acid

Give the mechanism of nitration of benzene.
